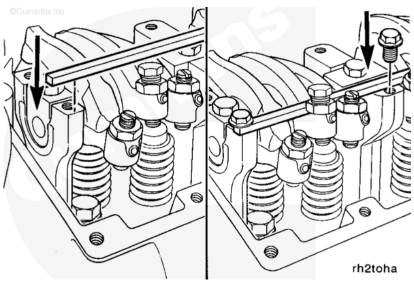
Trainings

Preparatory Steps

- Remove the air piping from the intake manifold.
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html)
- Remove the engine brakes, if equipped. Refer to Procedure 020-024 in Section 20. (/qs3/pubsys2/xml/en/procedures/35/35-020-024-tr.html)
- Remove the push rods and push tubes. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/35/35-004-014-tr.html)

Remove



Note : This action applies only to engines built with cast-iron supports.

Install a piece of 1/4-inch key stock 457 mm [18 in] long on top of the four front rocker lever assembly supports. Use four M10 - 1.50 x 25 flange head capscrews to secure the bar stock to the supports.

The capscrews on the two end supports fasten to the engine brake mounting holes on one side of the bar stock. The capscrews on the two center supports fasten to the engine brake mounting holes on the opposite side of the bar stock.

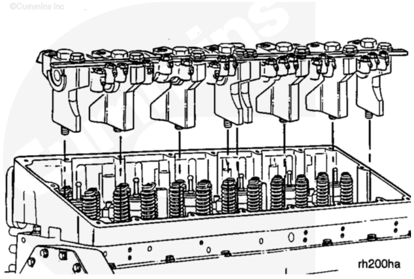
Loosen, but do **not** remove, the eight rocker shaft capscrews. The capscrews hold the rocker lever assemblies together.

Note : Engines with cast-iron supports: Grasp the bar stock and lift the front rocker lever assembly from the

engine. Do not lift the assemblies by grasping the rocker levers and shaft. The shaft and levers can lift out of the supports allowing the levers to slide off the shaft.

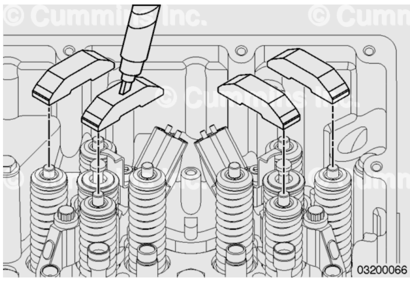
Note : Engines with aluminum supports: Grasp the assembly by the rocker levers and shaft and lift them from the engine. The shafts can not lift out of the supports.

Repeat the process to remove the rear rocker lever assembly.



Remove the crossheads.

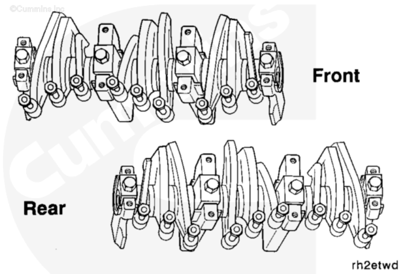
Note : Be sure to note the crosshead location so they can be installed in the same location.



Disassemble

Mark each rocker lever, shaft, and support with their relative position in the engine.

The rocker lever assemblies **must** be installed in the same position from which they were removed.

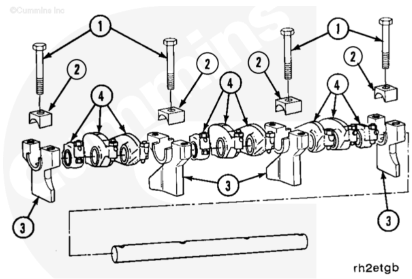


Remove the rocker shaft capscrews (1) and retaining blocks (2).

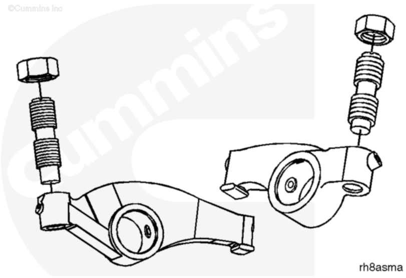
If the supports are cast-iron, lift the shaft and levers out of the supports (3).

If the supports are aluminum, slide the end supports and levers off the end of the shafts.

Remove the rocker levers (4) from both shaft assemblies.



Remove the locknuts and the adjusting screws from each rocker lever.



Clean

⚠ WARNING ⚠

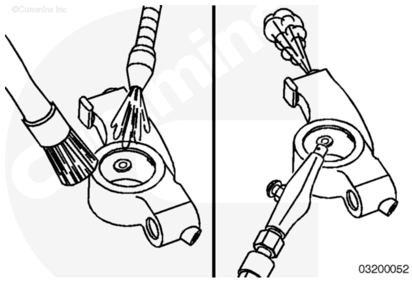
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

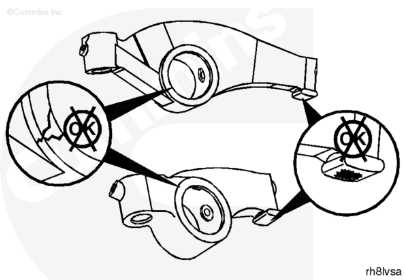
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use steam or solvent to clean the parts. Dry with compressed air.

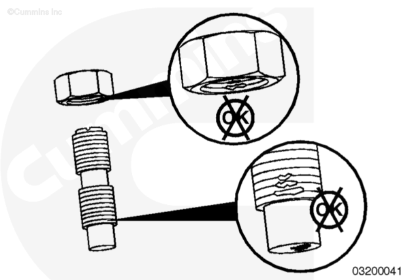
Be sure to blow out the oil passages.



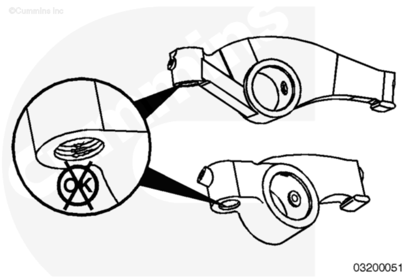
Inspect the rocker levers for cracks, excessive pitting, or unusual wear.



Inspect the adjusting screws and locknuts for damaged threads.

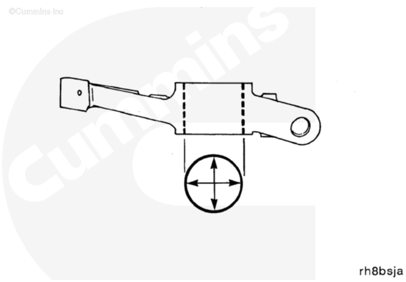


Inspect the adjusting screw threads in the rocker levers for damaged threads.



Measure each rocker lever bushing bore inside diameter.

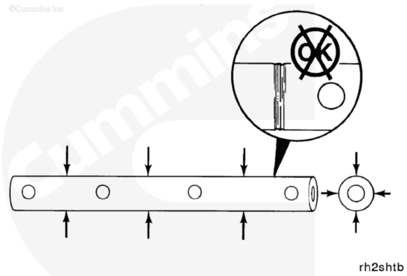
Rocker Lever Bushing Inside Diameter (Installed)		
mm		in
34.887	MIN	1.3735
34.990	MAX	1.3776



Inspect the rocker lever shafts for pitting, scoring, or other damage.

Measure each rocker lever shaft outside diameter.

Rocker Lever Shaft Outside Diameter		
mm		in
34.837	MIN	1.3715
34.864	MAX	1.3726

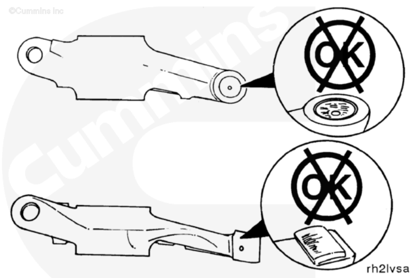


If worn or damaged parts are found, or the rocker lever bushings or shafts are **not** within the specifications given, the rocker lever assemblies **must** be rebuilt.

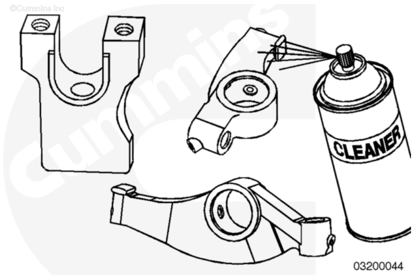
Inspect the sockets in the injector rocker levers for wear or damage.

Inspect the valve rocker lever pads for wear, cracks, or other damage.

If wear, cracks, or other damage is found, the rocker lever **must** be replaced.

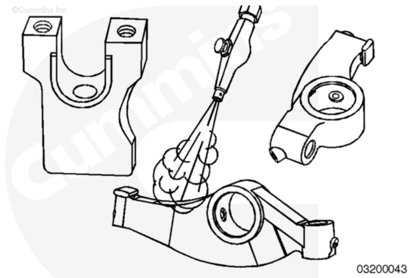


Use crack detection kit, Part Number 3375432, to inspect the rocker lever supports for cracks or damage.



⚠ WARNING ⚠

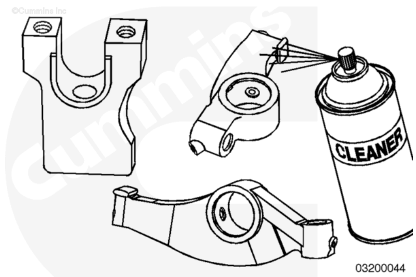
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Use crack detection cleaner, Part Number 3375433, to clean the rocker levers and shaft supports. Dry with compressed air.

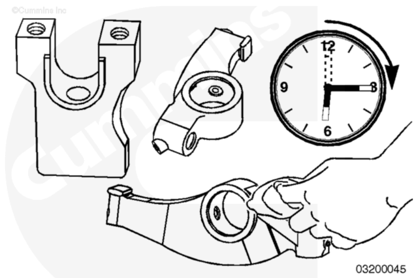
Use crack detection penetrant, Part Number 3375435, to spray the rocker levers and shaft supports.

Note : Do not dry with compressed air.

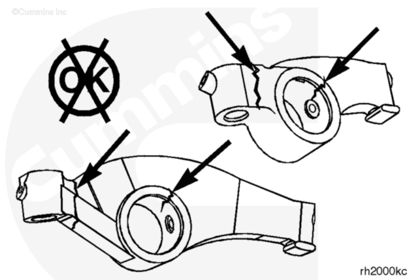


Allow the penetrant to dry for 15 minutes.

Remove the excess penetrant with a dry cloth.



Use crack detection developer, Part Number 3375434, to spray the rocker levers and shaft supports.



Inspect the levers and supports.

Cracks will appear as a solid, bright line.

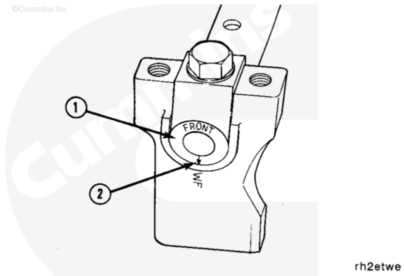
Cavitation in the casting will appear as a small round mark.

If cracks or cavitation are found, the part **must** be replaced.

Assemble

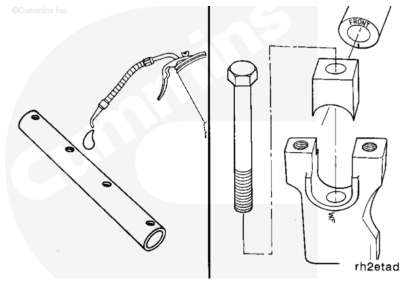
⚠ CAUTION ⚠

Make sure the rocker lever shafts are installed with the arrow on the end of the shafts pointing DOWN. When the arrow points up, cross-drilling to the rocker levers are orientated differently, which has an adverse affect to rocker lever oiling.



The rocker lever shafts are labeled front (or "F") and rear on the end of the shafts (1). The arrows on the shafts (2) **must** be pointed downward to maintain an oil flow to the rocker levers.

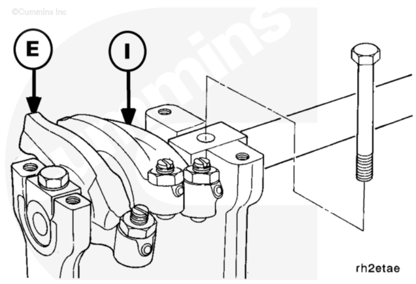
The shaft end supports are **not** interchangeable.



Use clean 15W-40 engine oil to lubricate both shafts.

On cast-iron supports, install the shaft into the front end support. Install the retaining block, washer, and capscrew through the support.

On aluminum supports, install the support by sliding it onto the end of the shaft. Install a washer and capscrew through the support.

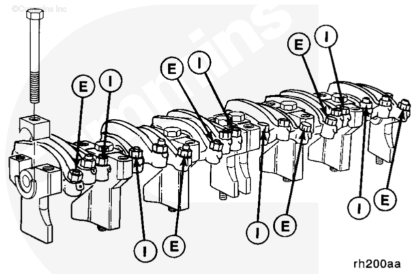


Install the rocker levers in the correct sequence, as shown.

Note : The rocker levers must be installed so they will be in the same position in the engine as when they were removed.

Install one of the two shaft center supports on the shaft.

On cast-iron supports, install the retaining block, washer, and capscrew through the center support.



Install the remaining levers and supports with the intake (I) and exhaust (E) valve levers in the correct position, as shown.

On cast-iron supports, install the remaining retaining blocks, washers, and capscrews.

On cast-iron supports **only**, install the key stock used to remove the rocker levers on top of the supports.

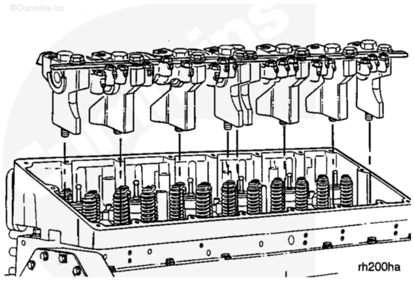
Install

The rocker lever assemblies **must** be installed in the engine so they are in the same position as they were removed from.

Install the assemblies on the engine.

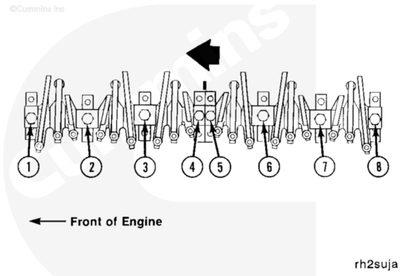
Note : Hand-tighten the mounting capscrews. The rocker lever side clearance must be adjusted before the capscrews are tightened to their final torque value.

Rocker Lever Installation Side Clearance		
mm		in
0.50	NOM	0.020



Push or use a hammer to tap the number 5 rocker lever support toward the front of the engine.

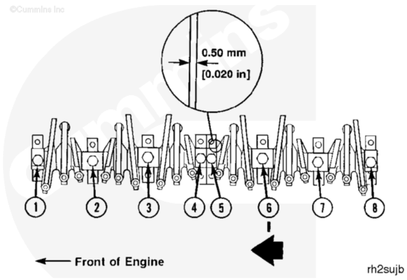
Tighten the mounting capscrews to 5 N•m [44 in-lb].



Install a 0.50 mm [0.020 in] feeler gauge between the number 5 support and the intake lever for the number 4 cylinder.

Push or use a hammer to tap the number 6 support toward the front of the engine.

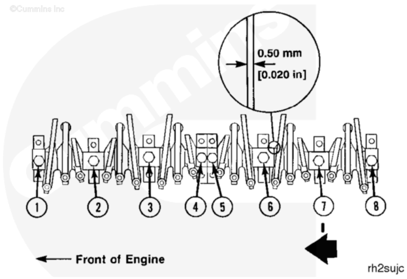
Tighten the mounting capscrews to 5 N•m [44 in-lb].



Install a 0.50 mm [0.020 in] feeler gauge between the number 6 support and the exhaust lever for the number 5 cylinder.

Push or use a hammer to tap the number 7 support toward the front of the engine.

Tighten the mounting capscrews to 5 N•m [44 in-lb].

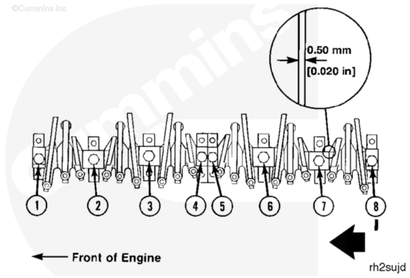


Install a 0.5 mm [0.020 in] feeler gauge between the number 7 support and the intake lever for the number 6 cylinder.

Push or use a hammer to tap the number 8 support toward the front of the engine.

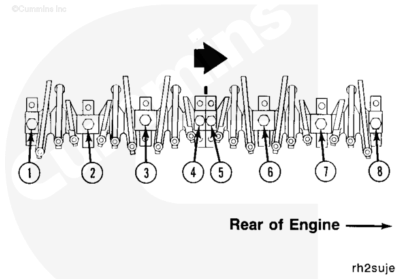
Tighten the number 5, 6, 7, and 8 support capscrews to their final torque value.

Torque Value: 122 n•m [90 ft-lb]



Push or use a hammer to tap the number 4 support toward the rear of the engine.

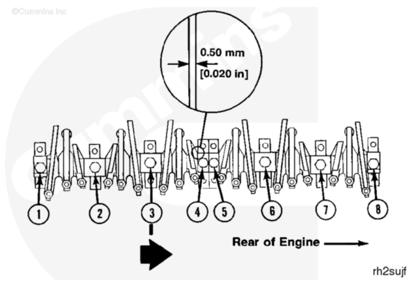
Tighten the mounting capscrews to 5 N•m [44 in-lb].



Install a 0.50 mm [0.020 in] feeler gauge between the number 4 support and the intake lever for the number 3 cylinder.

Push or use a hammer to tap the number 3 support toward the rear of the engine.

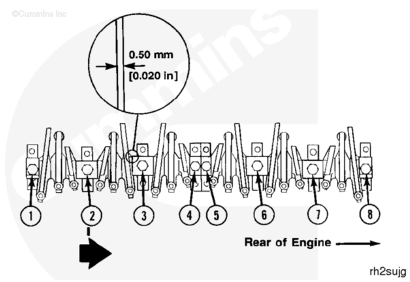
Tighten the mounting capscrews to 5 N•m [44 in-lb].



Install a 0.50 mm [0.020 in] feeler gauge between the number 3 support and the exhaust lever for the number 2 cylinder.

Push or use a hammer to tap the number 2 support toward the rear of the engine.

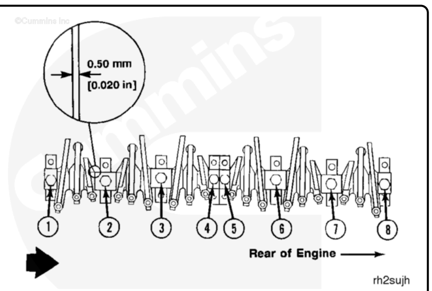
Tighten the mounting capscrews to 5 N•m [44 in-lb].



Install a 0.50 mm [0.020 in] feeler gauge between the number 2 support and the intake lever for the number 1 cylinder.

Push or use a hammer to tap the number 1 support toward the rear of the engine.

Tighten the number 1, 2, 3, and 4 support capscrews to their final torque value.

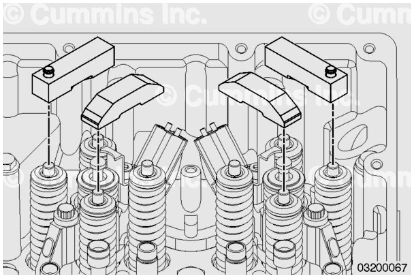


Torque Value: 122 n•m [90 ft-lb]

Check the front and rear assemblies for the correct clearance. Check the support capscrews for the correct torque value.

Install the valve crossheads.

Note : The crossheads must be installed in the same location from which they were removed.



Engines Equipped with Engine Brakes:

Install the six engine brake crossheads on the exhaust valves. Verify the actuator pins are facing the camshaft side of the engine.

Verify the crossheads are fully seated on the valve stems.

Finishing Steps

- Install the push rods and push tubes. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/35/35-004-014-tr.html)
- Adjust the valves and injectors. Refer to Procedure 003-004 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-004-tr.html)
- Install the engine brakes, if applicable. Refer to Procedure 020-024 in Section 20. (/qs3/pubsys2/xml/en/procedures/35/35-020-024-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html)
- Install the air piping to the intake manifold. Refer to the manufacturer's specifications for the correct torque value.
- Operate the engine and check for proper operation.